

Xincheng Huang

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RESEARCH STATEMENT

My research in Human-Computer Interaction develops immersive and intelligent AR/VR systems using neural scene representations (NeRF, Gaussian Splatting) and AI models (LLMs, VLMs). My thesis focuses on interactive and intelligent telepresence: efficient photorealistic capture, interactive shared surfaces and 3D object exchange, and AI-driven spatial understanding for adaptive collaboration. My work has been published at top Human-Computer Interaction and AR/VR venues such as UIST, CHI, IMWUT/UbiComp, and IEEE TVCG.

EDUCATION

Doctor of Philosophy in Computer Science Sept 2021 – Aug 2027
University of British Columbia, Vancouver, BC
Advisor: Dr. Robert Xiao

Master of Science in Computer Science and Engineering Aug 2019 – April 2021
University of Michigan, Ann Arbor, MI
Advisors: Dr. Nikola Banovic and Dr. Alanson Sample

Bachelor of Science in Computer Science Aug 2015 – May 2019
New York University Shanghai, China
Graduated with Magna Cum Laude, and a secondary major in Interactive Media Arts

RESEARCH EXPERIENCE

Student Researcher May 2026 – Oct 2026
Google. Waterloo, BC
Google XR. Mentors: Ruofei Du and Zhongwei Zhao

Graduate Research Assistant Sept 2021 – present
University of British Columbia, Vancouver, BC
X Lab. Advisor: Dr. Robert Xiao

Research Assistant Mar 2020 – April 2021
University of Michigan, Ann Arbor, MI
Computational HCI Lab. Advisors: Dr. Nikola Banovic and Dr. Alanson Sample

Research Assistant Jan 2020 – April 2020, Jan 2021 – April 2021
University of Michigan, Ann Arbor, MI
Secure Cloud Manufacturing Group. Advisor: Dr. Kira Barton

Undergraduate Research Assistant. Jan 2018 – Dec 2018
New York University, New York City, NY
NYU – Guggenheim Computer-based Art Conservation Initiative. Advisor: Prof. Deena Engel

PUBLICATIONS

Top-tier publications in HCI and AR/VR appear in ACM/IEEE conference proceedings such as CHI, UIST, Ubicomp, IEEEVR, and ISMAR. Ubicomp has a hybrid structure with journal papers appearing in IMWUT. IEEEVR and ISMAR accept top submissions (~10%) to the journal-track and publish them to IEEE TVCG.

Conference Proceedings (Refereed)

[C.5] Hongyu Zhou, Xincheng Huang, Winston Wijaya, Yi Fei Cheng, David Lindlbauer, Eduardo Velloso, Andrea Bianchi, Zhanna Sarsenbayeva, and Anusha Withana. 2026. One Body, Two Minds: Alternating VR Perspective During Remote Teleoperation of Supernumerary Limbs. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 20 pages. <https://doi.org/10.1145/3772318.3791433>. (Acceptance rate: 25.3%)

[C.4] Xu Fan, Xincheng Huang, and Robert Xiao. 2025. *TangiAR*: Markerless Tangible Input for Immersive Augmented Reality with Everyday Objects. In *31st ACM Symposium on Virtual Reality Software and Technology (VRST '25)*, November 12–14, 2025, Montreal, QC, Canada. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3756884.3766028>. (Acceptance rate: 27%)

[C.3] Xincheng Huang, Dieter Frehlich, Ziyi Xia, Peyman Gholami, and Robert Xiao. 2025. GaussianNexus: Room-Scale Real-Time AR/VR Telepresence with Gaussian Splatting. In *The 38th Annual ACM Symposium on User Interface Software and Technology (UIST'25)*. September 28–October 1, 2025, Busan, Korea. ACM, New York, NY, USA, 18 pages. <https://doi.org/10.1145/3746059.3747693>. (Acceptance rate: 22%)

[C.2] Ziyi Xia, Xincheng Huang, Sidney S Fels, and Robert Xiao. 2025. HaloTouch: Using IR Multi-Path Interference to Support Touch Interactions with General Surfaces. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 548, 1–17. <https://doi.org/10.1145/3706598.3714179>. (Acceptance rate: 24.9%)

[C.1] Xincheng Huang, Michael Yin, Ziyi Xia, Robert Xiao. 2024. VirtualNexus: Enhancing 360-Degree Video AR/VR Collaboration with Environment Cutouts and Virtual Replicas. In *The 37th Annual ACM Symposium on User Interface Software and Technology (UIST '24)*, October 13–16, 2024, Pittsburgh, PA, USA. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3654777.3676377>. (Acceptance rate: 24%)

Journal Articles (Refereed)

[J.5] Ritik Vatsal, Xincheng Huang, Robert Xiao, "Look2React: Making VR NPCs Come Alive with Dynamic Vision-Guided Reactions," in *IEEE Transactions on Visualization and Computer Graphics*, doi: [10.1109/TVCG.2026.3679121](https://doi.org/10.1109/TVCG.2026.3679121). (Acceptance rate: 20.6%)

[J.4] Bu Li, Xincheng Huang, and Robert Xiao. 2025. VibRing: A Wearable Vibroacoustic Sensor for Single-Handed Gesture Recognition. *Proc. ACM Hum.-Comput. Interact.* 9, 4, Article EICS006 (June 2025), 25 pages. <https://doi.org/10.1145/3733052>.

[J.3] Xincheng Huang and Robert Xiao. 2023. SurfShare: Lightweight Spatially Consistent Physical Surface and Virtual Replica Sharing with Head-mounted Mixed-Reality. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 7, 4, Article 162 (December 2023), 24 pages. <https://doi.org/10.1145/3631418>.

[J.2] Xincheng Huang, James Riddell, and Robert Xiao. 2023. “Virtual Reality Telepresence: 360-Degree Video Streaming with Edge-Compute Assisted Static Foveated Compression”, in *IEEE Transactions on Visualization and Computer Graphics*, doi: [10.1109/TVCG.2023.3320255](https://doi.org/10.1109/TVCG.2023.3320255). (Acceptance rate: 20%)

[J.1] Xincheng Huang, Keylonnie L. Miller, Alanson P. Sample, and Nikola Banovic. 2023. StructureSense: Inferring Constructive Assembly Structures from User Behaviors. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 6, 4, Article 204 (December 2022), 25 pages. <https://doi.org/10.1145/3570343>.

Posters/Demo (Reviewed/Juried)

[PD.1] Dieter Frehlich, Xincheng Huang, Junkai Ding, Robert Xiao. 2025. MultiSphere: Latency Optimized Multi-User 360° VR Telepresence with Edge-Assisted Viewport Adaptive IPv6 Multicast. In *31st ACM Symposium on Virtual Reality Software and Technology (VRST '25)*, November 12–14, 2025, Montreal, QC, Canada. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3756884.3770539>

[PD.2] Xincheng Huang, Dieter Frehlich, Ziyi Xia, Peyman Gholami, and Robert Xiao. 2026. Demonstrating GaussianNexus: Room-Scale Real-Time AR/VR Telepresence with Gaussian Splatting. In *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*. ACM, New York, NY, USA, Article 836, 1–6. <https://doi.org/10.1145/3772363.3799125>

Patents

[P.1] Xincheng Huang, Dieter Frehlich, Ziyi Xia, Peyman Gholami, Robert Xiao. (2025). Mixed Reality System for Real-Time AR/VR Telepresence. Patent Application filed September, 2025. Patent pending.

[P.2] Xincheng Huang, James Riddell, Ailin Saggau-Lyons, Robert Xiao. System and Apparatus for Video Display on a Portable Display Device. **US Patent** 12,604,019, filed October 13th 2023 and granted April 14th 2026.

TEACHING AND MENTORING

Graduate Teaching Assistant <i>University of British Columbia, Vancouver, BC</i> CPSC 554X – Machine Learning and Signal Processing. Duties: grading and responding to student questions.	Sept 2022 – Dec 2022
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Learning Assistant <i>New York University Shanghai, Shanghai, China</i> CSCI-SHU 101 Introduction to Computer Science. Duties: holding tutoring office hours, conducting review sessions, and facilitating class activities. Received award for “Excellent Tutoring” and “Most Appointed Office Hour”.	Feb 2019 – May 2019
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Mentoring

Michael Dickinson, undergrad researcher at *UBC with WLIURA* award
Michael (He) Wang, undergrad researcher at *UBC with SURA* award
Wenbo Wang, undergrad researcher at *UBC with WLIURA* award
Danning Hao, undergrad researcher at *UBC with WLIURA* award
Ritik Vatsal, master’s student at *UBC*, currently a lab member at *UBC*
Dieter Frehlich, undergrad at *UBC*, then PhD student at *UBC*
Junkai (Kelvin) Ding, undergrad at *UBC*, then master’s at *Northeastern University*

James Riddell, undergrad at *UBC*, then master's at *the University of Waterloo*
Keylonnie Miller, undergrad at *Umich*, then *Facebook*

INVITED PRESENTATIONS/DEMOS

“Demonstrating GaussianNexus: Room-Scale Real-Time AR/VR Telepresence with Gaussian Splatting”. Interactive demo at CHI 2026, Barcelona, Spain. April 2026.

“Together from Afar: Toward Immersive, Interactive, and Intelligent Spatial Telepresence”. Online Guest Lecture at Oxford Brookes University, U.K. Hosted by Prof. Fridolin Wild, February 24th, 2026.

“Together from Afar: Toward More Natural Spatial Telepresence”. Research Talk at the University of Texas at Dallas, USA. Hosted by Prof. Jin Ryong Kim, October 31st, 2025.

“Together from Afar: Toward More Natural Spatial Telepresence”. Research Talk at Tsinghua University, Beijing, China. Hosted by Prof. Xin Yi, October 20th, 2025.

“VirtualNexus: Enhancing 360-Degree Video AR/VR Collaboration with Environment Cutouts and Virtual Replicas”. Live demo at the 2nd *Beyond Connectivity Research Summit* by Rogers Communications Inc. Vancouver, Canada, July 2024.

“Virtual Reality Telepresence: 360-Degree Video Streaming with Edge-Compute Assisted Static Foveated Compression”. Live demo at the 1st *Beyond Connectivity Research Summit* by Rogers Communications Inc. Vancouver, Canada, June 2023.

ACADEMIC SERVICE

Program Committee: CHI 2025 Late-Breaking Work, ISS 2026

Session Chair: UIST 2025, CHI 2026

External Reviewer (65 papers in various journals and conferences)

UIST 2023, 2025, CHI 2023-2026, IMWUT 2024-2025, IEEEVR 2025-2026, ISMAR 2024-2025, TVCG 2025, IHCS 2025, DIS 2026, ISS 2026, SUI 2024-2025, VRST 2024-2025, EICS 2025, CHI Play 2024-2026, IMX 2025
Received Excellent Review Recognitions for CHI 2025-2026, UIST 2025-2026, DIS 2026, CHI Play 2024, 2026

Student Volunteer: ISS 2024

Presentation Judge: UBC CS Student Research Conference

AWARDS

MITACS Accelerate 2023-2024
Conducting Project *Rich, Immersive AR/VR communication* in collaboration with *Rogers Communications Canada Inc.* with a [Mitacs Accelerate](#) award with 30000 CAD.

Latin Award, Magna Cum Laude 2019
Awarded to top 15% of the graduated class at New York University Shanghai

Dean's List for Academic Year

2015 – 2016, 2016 – 2017, 2017 – 2018, 2018 – 2019

*Awarded to top 30% for each academic year at New York University Shanghai***LEADERSHIP****ENACTUS NYU Shanghai, Vice President**

2016 - 2017

Led the NYU Shanghai's branch of [ENACTUS](#), a world-wide social entrepreneurial student organization. Participated in the user interviews, product design, and prototyping for our project: "A Third Eye: A Digital Blind Crutch". Won the *First prize* and the *Best technology innovation award* in ENACTUS social innovation competition of East China, and an 80000 CNY (~12000 USD) grant from the Chinese Charity Association (Shenzhen).

SKILLS**Programming:** Experienced with C#, Python, Java, C/C++, OpenGL, HLSL.**AR/VR Development:** Proficient in Unity, Mixed Reality Tool Kit, and XR Interaction Toolkit. Experienced with Meta Quest and Microsoft HoloLens development.**Machine Learning & AI:** Experienced with OpenCV and PyTorch.**Prototyping:** Experienced with Arduino, Raspberry Pi, Processing.js, and p5.js.**Design Software:** Experienced with Adobe Photoshop, Premiere Pro, and Illustrator. Familiar with Maya, Cinema4D, and Blender.